

ZERAVIA

Training & Internship Program

Machine Learning with Python

Mini Project Report

Customer Purchase Prediction & Behaviour Analysis

Student Name: Anwesha Singha
Dataset: Supermarket Sales Dataset
Observations: 1000
Original Variables: 17
Primary ML Task: Binary Classification
Secondary ML Task: Sales Regression

A beginner-level end-to-end machine learning workflow for a retail client

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1 Project Overview

This project develops a complete machine learning workflow for a "Supermarket client". It follows the structure: problem definition, Python setup, data loading, Python fundamentals, NumPy/Pandas, data cleaning and transformation, exploratory data analysis, visualisation, statistics, preprocessing, train-test split, Linear Regression where applicable, Logistic Regression, evaluation, business insights and client recommendations.

2 Data Collection

This Supermarket Sales Dataset is taken from "Kaggle".

This captures detailed transaction records from a supermarket chain in Myanmar across three cities: Yangon, Naypyitaw, and Mandalay. It provides a comprehensive view of sales activities, customer demographics, and payment methods from January to March 2019. The dataset contains 1000 supermarket transactions and 17 original variables containing branch, city, customer type, gender, product line, unit price, quantity, tax, sales, date, time, payment, cost of goods sold, gross margin, gross income and rating.

3 Problem definition Objective

3.1 Business Problem

The supermarket has historical transaction records and wants to understand purchasing behaviour. Management is particularly interested in distinguishing 'Member customers' from 'Normal customers' and using transaction-level information to support customer engagement and campaign targeting.

3.2 Prediction Objective

The primary target is **Customer type**, represented as

$$Y = \begin{cases} 1, & \text{Member} \\ 0, & \text{Normal} \end{cases}$$

This is clearly a supervised binary classification problem because the historical target label is observed. A secondary continuous task predicts **Sales**. This allows Linear Regression to be demonstrated. Linear Regression is not appropriate for the binary customer-type target, so it is treated as a separate modelling objective.

4 Python Fundamentals setup

The workflow uses lists for numerical and categorical feature names, dictionaries for storing calculated results, sets for checking unique categories, and **for** loops for repeated summaries and IQR calculations. Conditional logic can be used for simple validation rules.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

5 Data Loading

The dataset is loaded using Pandas:

```
df = pd.read_csv("SuperMarket_Analysis(1).csv")
print(df.head())
print(df.info())
```

The dataset has **1000 observations and 17 original columns**. There are **0 missing values and 0 duplicate rows**. Customer type contains 565 Member transactions and 435 Normal transactions.

6 Data Cleaning and Transformation

The Date variable was converted to datetime format. Three temporal features were created:

- Month: month name;
- Day: day of week;
- Hour: hour extracted from transaction time.

The original draft code referred to a variable called **Total**; the supplied dataset actually contains **Sales**, so **Sales** is used in the final analysis.

```
df["Date"] = pd.to_datetime(df["Date"])
df["Month"] = df["Date"].dt.month_name()
df["Day"] = df["Date"].dt.day_name()
df["Hour"] = pd.to_datetime(
    df["Time"], format="%I:%M:%S%p"
).dt.hour
```

7 NumPy and Pandas Analysis

NumPy was used for array conversion and numerical calculations. For Sales, the mean is **322.97**, the median is **253.85**, and the standard deviation is **245.89**. Pandas was used for loading, inspection, missing-value checks, duplicate checks, grouping, aggregation, filtering and feature engineering.

8 Exploratory Data Analysis

8.1 Selected Numerical Summary

Variable	Mean	SD	Min	Median	Max
Unit price	55.67	26.49	10.08	55.23	99.96
Quantity	5.51	2.92	1	5	10
Sales	322.97	245.89	10.68	253.85	1042.65
Rating	6.97	1.72	4.0	7.0	10.0

Table 1: Selected descriptive statistics.

8.2 Meaningful EDA Insights

1. **Member customers are more frequent:** 565 Member transactions versus 435 Normal transactions.
2. **Member transactions have higher average sales:** approximately 335.74 versus 306.37.
3. **Female transactions have greater observed sales:** 571 transactions and approximately 194671.84 total sales, versus 429 transactions for males.
4. **Branch performance is relatively balanced:** Giza has the highest total sales, while Alex and Cairo are close behind.
5. **Product lines are broadly distributed:** Food and beverages has the largest transaction count, while Sports and travel has very strong sales.
6. **Payment methods are all equally important:** Cash, Ewallet and Credit card have 344, 345 and 311 transactions respectively.
7. **Saturday is the strongest day by total sales:** approximately 56120.81.
8. **19:00 is the strongest observed sales hour:** approximately 39699.51 in total sales.

9 Visualisation

This project provides distributions, category comparisons, temporal analysis, correlation analysis and model evaluation plots.

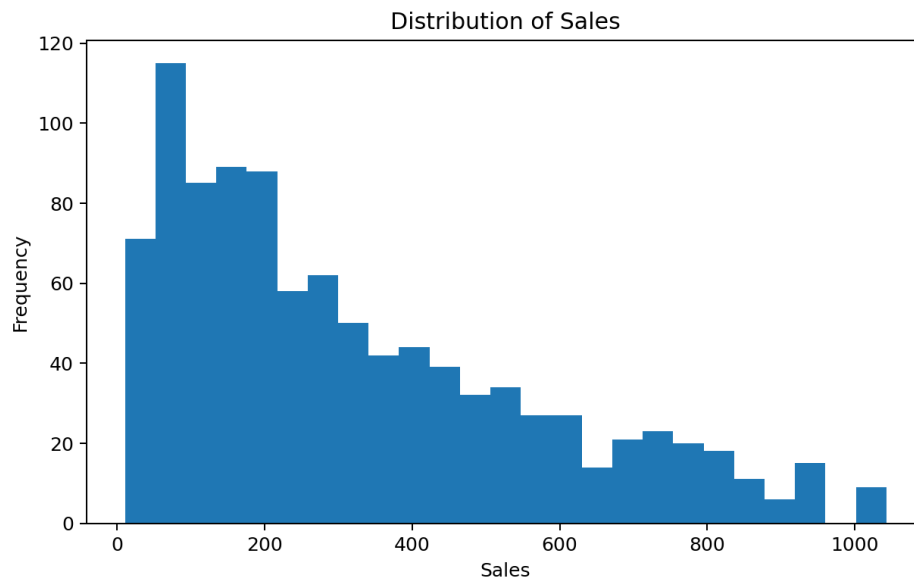


Figure 1: Distribution of Sales.

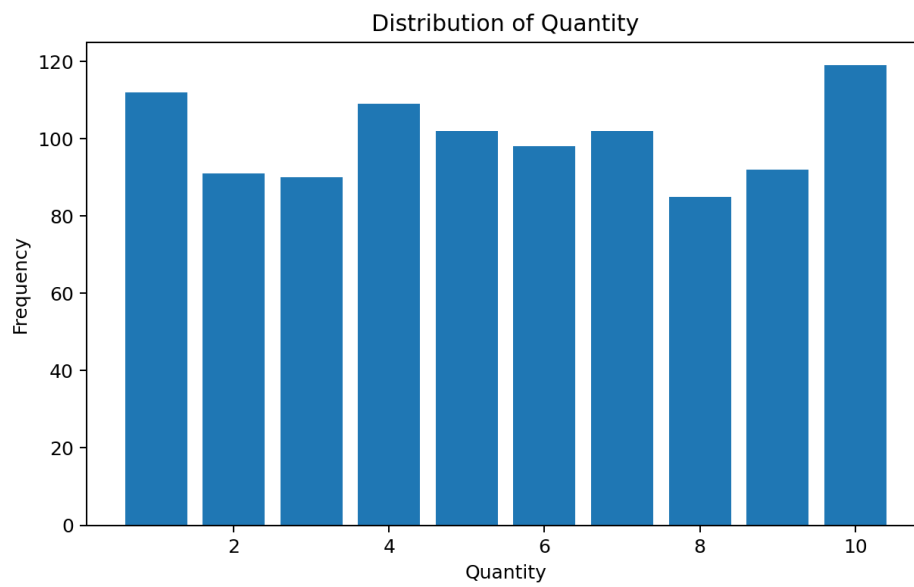


Figure 2: Distribution of Quantity.

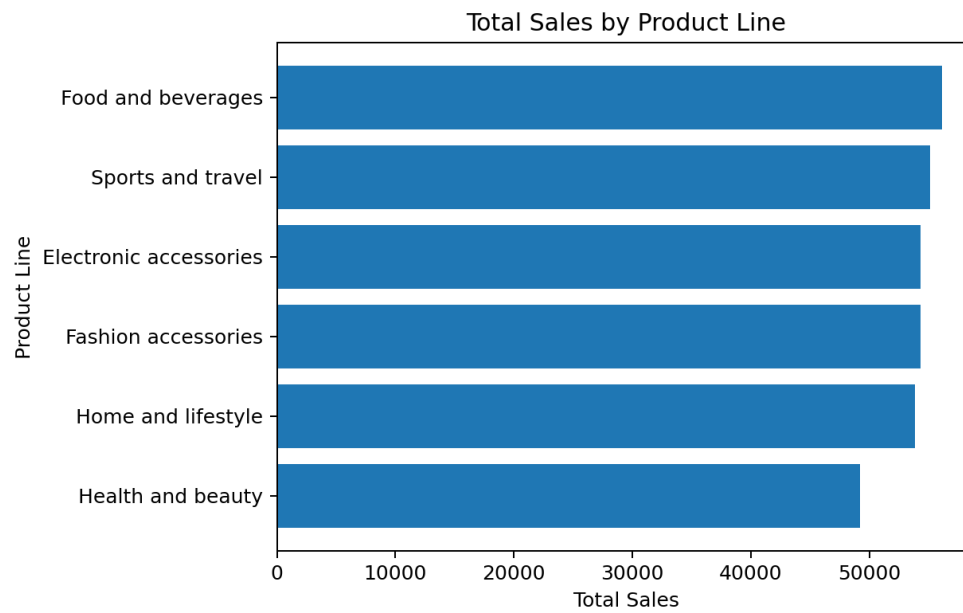


Figure 3: Total sales by product line.

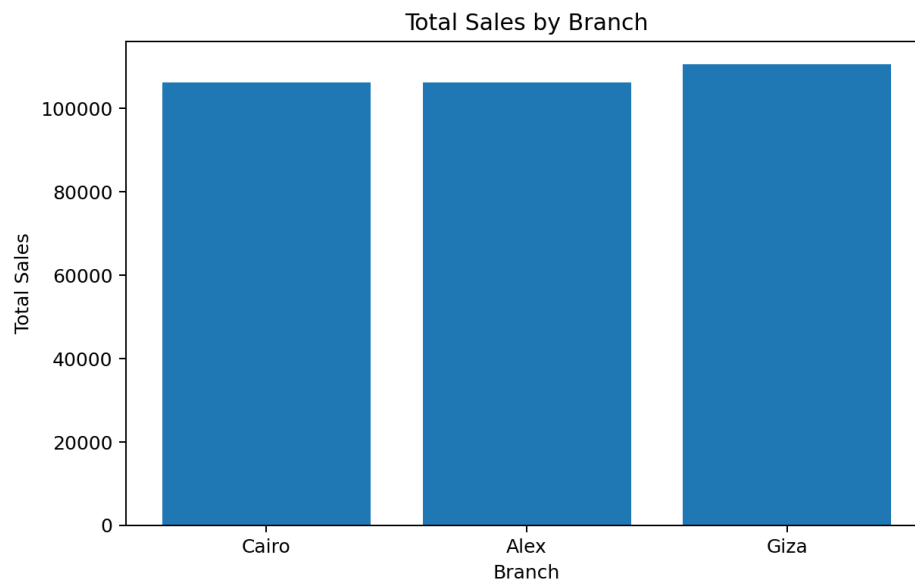


Figure 4: Total sales by branch.

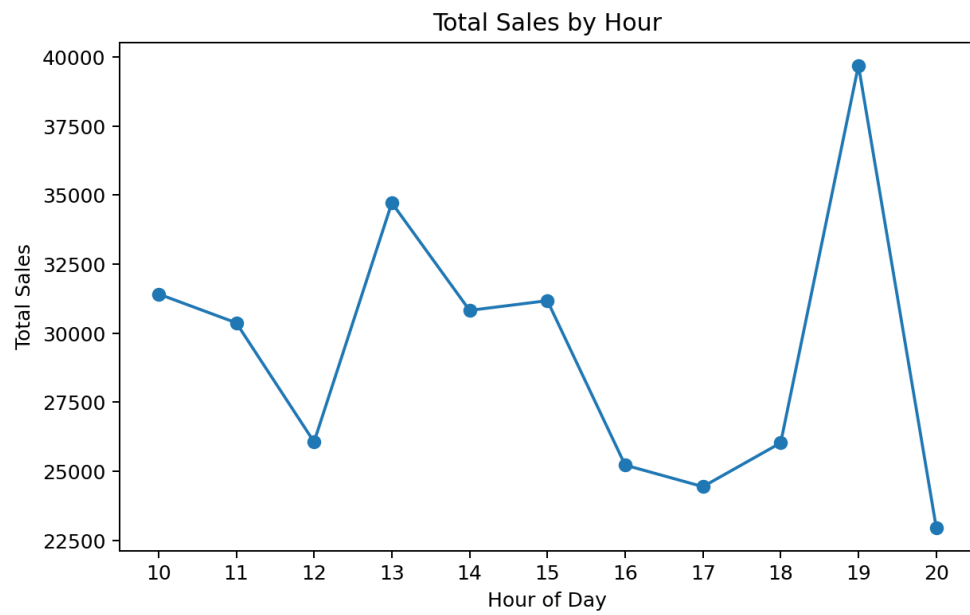


Figure 5: Total sales by hour.

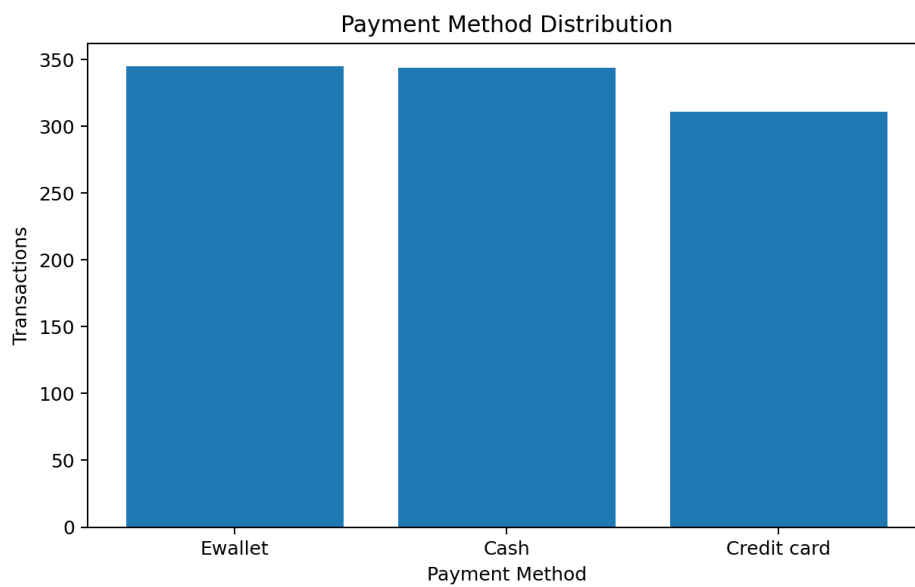


Figure 6: Payment method distribution.

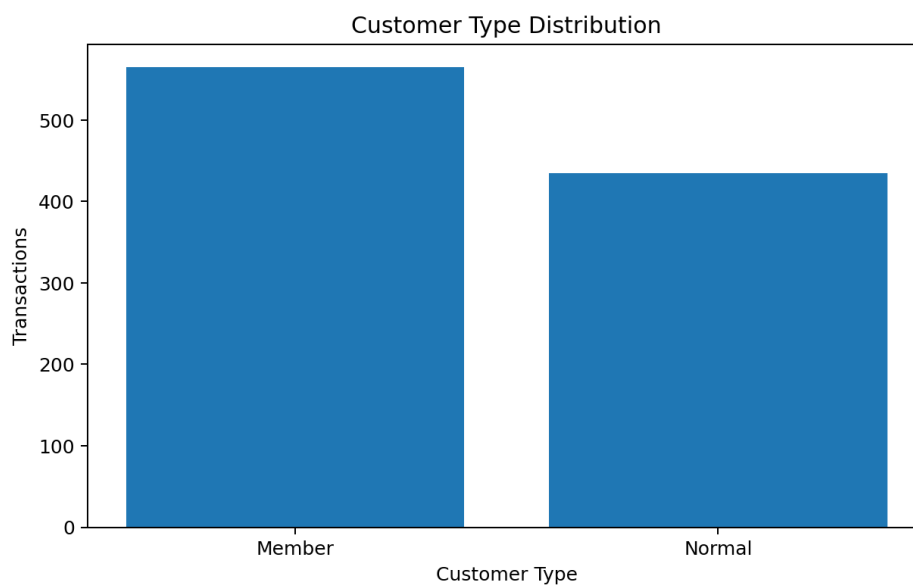


Figure 7: Customer type distribution.

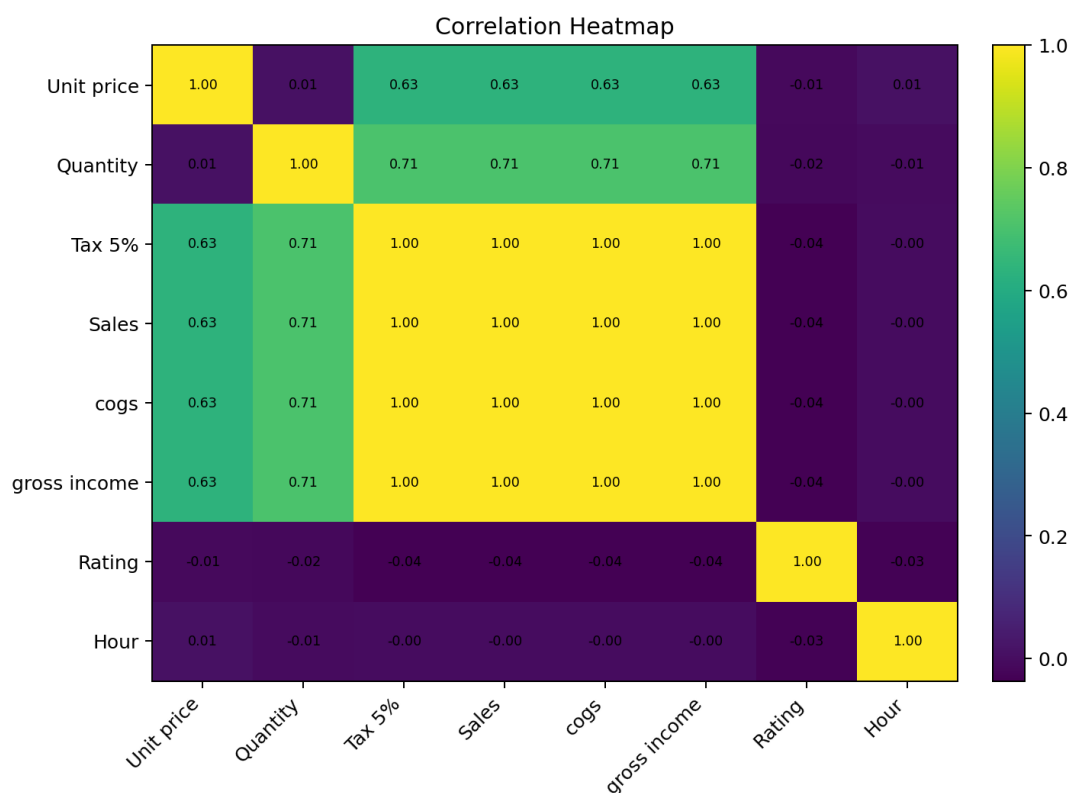


Figure 8: Correlation heatmap of numerical variables.

10 Statistical Thinking and IQR Outliers

The IQR rule was used:

$$IQR = Q_3 - Q_1, \quad L = Q_1 - 1.5IQR, \quad U = Q_3 + 1.5IQR.$$

Variable	Potential outliers
Unit price	0
Quantity	0
Tax 5%	9
Sales	9
cogs	9
gross income	9
Rating	0
Hour	0

Table 2: IQR-based potential outliers.

Nine potential outliers occur in Sales and correspondingly in Tax 5%, cogs and gross income because these transaction-level financial variables are mathematically related. They were not automatically deleted because some may represent genuine high-value transactions.

11 Definition of the ML Problem

11.1 Primary Classification

- Learning type: supervised learning.
- Problem type: binary classification.
- Target: Member versus Normal.
- Output: predicted class and membership probability.
- Business use: customer segmentation and loyalty engagement.

11.2 Secondary Regression

Sales is continuous, so Linear Regression can estimate transaction sales from unit price, quantity, rating, time and categorical transaction characteristics.

12 Data Preprocessing

Numerical predictors were standardized using

$$z = \frac{x - \mu}{\sigma}.$$

Categorical variables were one-hot encoded. A `ColumnTransformer` and `Pipeline` were used so preprocessing is fitted within the training workflow.

```
preprocessor = ColumnTransformer([
    ("num", StandardScaler(), num_features),
    ("cat", OneHotEncoder(handle_unknown="ignore"),
     cat_features)
])

model = Pipeline([
```

```

("preprocessor", preprocessor),
("classifier", LogisticRegression(max_iter=1000))
])

```

13 Train-Test Split

The classification data were divided into 80% training and 20% testing observations using random state 42 and stratification. Thus 800 observations were used for training and 200 for testing. The test set remained unseen during model fitting.

14 Linear Regression

Linear Regression was fitted to predict Sales using unit price, quantity, rating, hour, branch, city, customer type, gender, product line, payment, month and day.

Metric	Test value
MAE	59.78
MSE	6417.83
RMSE	80.11
R^2	0.901

Table 3: Linear Regression performance for Sales prediction.

The R^2 value indicates that the model explains a substantial proportion of test-set Sales variation, while MAE describes the average absolute prediction error. This is a predictive baseline, not a causal model.

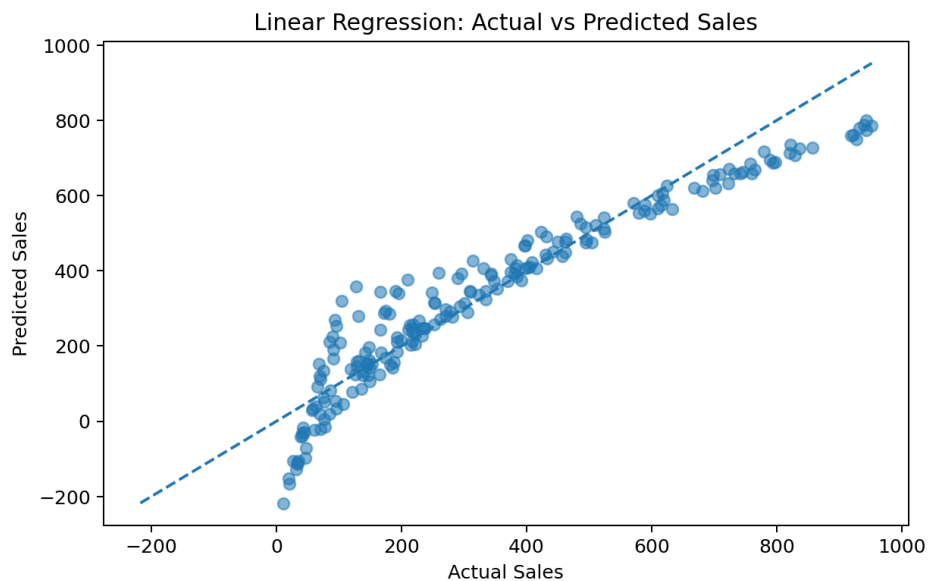


Figure 9: Actual versus predicted Sales.

15 Logistic Regression

Logistic Regression was used for the Member-versus-Normal classification problem. The test results were:

Metric	Test value
Accuracy	0.565
ROC-AUC	0.542
Normal precision	0.500
Normal recall	0.310
Normal F1	0.383
Member precision	0.589
Member recall	0.761
Member F1	0.664

Table 4: Logistic Regression classification performance.

The model achieved 0.565 PCT accuracy. Member recall was 0.761 PCT , while Normal recall was 0.310 PCT . AUC of 0.542 indicates modest discrimination on this test split. Consequently, the model should be used as a

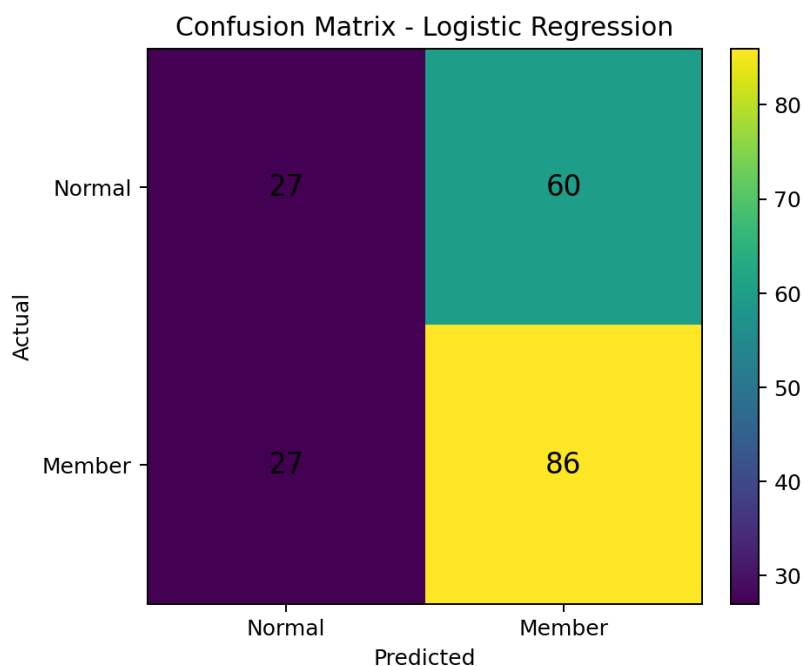


Figure 10: Confusion matrix.

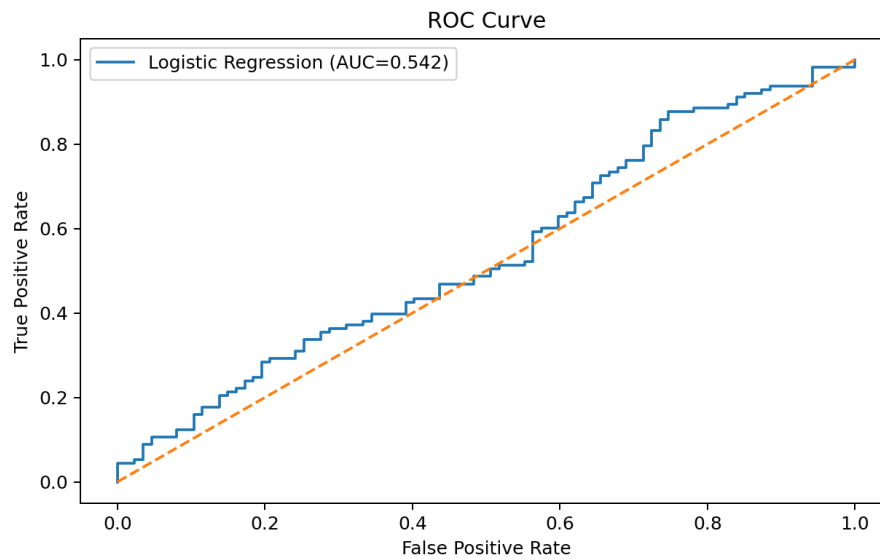


Figure 11: ROC curve.

16 Model Evaluation

16.1 Strengths

- Reproducible preprocessing and modelling pipeline.
- Appropriate handling of numerical and categorical variables.
- Class predictions and probabilities are available.
- Descriptive analytics are connected to predictive modelling.

16.2 Weaknesses

- Logistic Regression has limited predictive discrimination on the test split.
- The dataset may not contain strong direct indicators of membership.
- Several financial variables are mathematically related.
- The dataset contains only 1000 transactions and may not generalise to other settings.

17 Business Insights

The observed data show higher average sales among Members, relatively balanced branch performance, a larger observed sales contribution from female transactions, broad product-line activity, substantial use of all three payment methods, strong Saturday sales and a notable sales peak around 19:00. These patterns can inform staffing, promotions and loyalty campaigns.

18 Client Recommendations

1. **Strengthen loyalty conversion:** identify Normal customers with high predicted membership probability and offer membership benefits.

2. **Focus activity around high-sales periods:** provide additional promotional and operational support around peak hours, especially around 19:00.
3. **Use Saturday campaigns:** test loyalty and cross-selling campaigns on the strongest observed sales day.
4. **Maintain payment flexibility:** continue supporting Cash, Ewallet and Credit Card.
5. **Develop product-line cross-selling:** use high-volume categories such as Food and beverages for complementary offers.
6. **Use branch-level monitoring:** treat the stronger-performing Giza branch as a benchmark for operational comparison.
7. **Collect richer customer-level variables:** future models could use visit frequency, purchase history, tenure, campaign response and previous membership interactions.

19 Conclusion

This project implements an end-to-end machine learning workflow for a supermarket client. Data were inspected and cleaned, temporal features were engineered, descriptive statistics and IQR outlier analysis were performed, and more than six visualisations were produced.

Linear Regression provided a Sales-prediction baseline with an R^2 of 0.901. Logistic Regression achieved 0.565 PCT accuracy and 0.542 $ROC - AUC$ for Member – versus – Normal classification. The main practical conclusion is that transaction variables are useful for understanding

20 Limitations and Future Work

- Validate performance with cross-validation and, ideally, a later external test period.
- Investigate probability calibration and business-specific classification thresholds.
- Consider regularisation and feature selection because of correlated financial variables.
- Add longitudinal customer history and engagement variables.
- Compare with tree-based models when those methods are introduced in the curriculum.